ParkeSafe
FIND. BOOK. PARK SAFE.

ParkSafe

Smart Vehicle Parking & Online Slot Booking Application

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1. Introduction & Project Background

The Urban Parking Problem

- City roads are getting crowded with more cars and bikes every day.
- Drivers waste a lot of time searching for empty parking spaces.
- Traffic jams increase because vehicles keep looking for open slots.

The Shift to Digital Booking

- Traditional paper tickets are slow and easy to lose.
- Pre-booking parking slots online removes stress before a journey.
- Customers can check real-time slot availability on their phones.

The ParkSafe Vision

- A complete mobile app built with Flutter and Supabase cloud.
- AI helps find parking locations likely to have free slots.
- Fast QR code scanning for quick gate check-in and check-out.

2. Problem Statement: Urban Parking Crises

Time & Fuel Wastage

- Drivers spend 15 to 20 minutes just looking for a parking spot.
- This searching creates up to 30% of traffic jams in busy areas.

Lack of Real-Time Info

- People drive to parking places without knowing if slots are full.
- No clear information for Cars, Bikes, or EV Charging stations.

Paper Ticket Problems

- Lost paper tickets cause long arguments and gate delays.
- Extra time charges are hard to track without a digital timer.

Security & Verification

- Hard to verify ticket ownership without digital records.
- Drivers get no reminders when their parking time is ending.

3. Existing Problems vs. ParkSafe Solution

Traditional Manual Parking

• Blind Arrival: Drivers arrive and hope to find an open spot.

• Slow Paper Tickets: Paper slips cause long lines at entrance gates.

• No Floor Visibility: Managers cannot see live slot counts per floor.

• Untracked Overtime: Vehicles staying late are rarely charged correctly.

ParkSafe Smart Solution

• Guaranteed Pre-Booking: Reserve parking slots before leaving home.

• Digital QR Pass: Fast QR code scanning on smartphone.

• Multi-Floor Grid: Clear view of open slots floor by floor.

• Automatic Overtime: Late fees calculated automatically upon exit.

4. Proposed Solution: The ParkSafe Ecosystem

Easy Pre-Booking

- Book parking slots in advance for specific dates and times.
- Filter slots for Cars, Bikes, or EV Charging stations.
- View available floor slots on a single simple screen.

AI Parking Recommender

- Uses GPS to measure exact distance to nearby locations.
- Checks real-time slot availability to suggest best options.
- Fast Python FastAPI backend calculates top recommendations.

QR Gatekeeper & Alerts

- Instant QR code scanning for entry and check-out.
- Automatic overtime calculation if a driver leaves late.
- Phone push notifications for check-in and check-out reminders.

5. Key Objectives & Project Scope

Main Technical Goals

- No Double Booking: Database blocks two users from booking the same slot at the same time.
- Multi-Floor Support: Manage parking slots across multiple floor levels easily.
- Fast Cloud Sync: Instant update between Customer App and Admin Panel.
- Multi-Platform: Runs smoothly on Android, iOS, and Web browsers.

Project Scope & Boundaries

- Customer App: Location search, AI recommendation, slot booking, QR ticket.
- Admin Panel: Add locations, set floor slots, QR scanner, view live earnings.
- Python Backend: AI recommendation backend and automated notification worker.

6. Key Features & Capabilities Overview

Multi-Floor Setup

• Set custom floor layouts and slot limits for Car, Bike, and EV.

AI Recommender

• Ranks locations based on short distance and available free slots.

Overlap Protection

• Database checks time slots so two people cannot book the same slot.

QR Pass Scanner

• Quick camera scanning at gates with early check-in protection.

Push Notifications

• Automated reminders 15 minutes before check-in and check-out.

Overtime & Refunds

• Automatic late fee calculation and easy customer refund requests.

7. Customer Application Features

Search & Location Discovery

- Simple user registration and secure login.
- Search nearby parking places with Google Maps directions.
- View facilities like 24/7 CCTV, EV Charging, and Valet Parking.
- Get AI suggestions for nearest available parking spots.

Slot Reservation & Digital Pass

- Select booking date, start time, and end time.
- Filter slots for Cars, Bikes, or EV Charging stations.
- Choose floor and slot number on an interactive visual grid.
- Get an instant QR ticket pass saved under My Bookings.

8. Admin Panel Operational Suite Features

Facility & Slot Configuration

- Add or update parking locations with Google Maps links.
- Set total floors (Floor 1, Floor 2, Basement) and slot counts.
- Set separate slot limits for Cars, Bikes, and EV vehicles.
- View live dashboard with total earnings and today's bookings.

Gate Operation & Refunds

- Fast camera QR scanner to check vehicles in and out.
- Search by vehicle number plate if QR scanning fails.
- Blocks check-in if customer arrives before start time.
- Review and approve or reject customer refund requests.

9. Multi-Floor Parking & Slot Management

Admin Multi-Floor Setup

- Flexible Floor Setup: Each location can have different number of floors.
- Independent Floor Limits: E.g., Floor 1 (20 slots), Floor 2 (25 slots).
- Vehicle Allocation: Specify Car, Bike, and EV slots for each floor.
- Auto Slot Generator: System creates slot codes like C-01 or E-01 automatically.

Customer Floor Layout

- Single-Page Screen: All floors displayed together on one screen.
- Clear Floor Headers: Sections labeled Floor 1, Floor 2, Basement 1.
- Live Available Counters: Displays free slots available on each floor.
- Clear Ticket Details: Ticket shows exact location e.g., Floor 2 -> Slot C-24.

10. Slot Booking & Real-Time Availability

Real-Time Database Sync

- System checks active bookings before confirming a slot reservation.
- Visual Color Badges: Available slots show Green, Booked slots show Red.
- Instant Update: Cancelled bookings immediately open slots for others.
- Cloud Synchronization: Changes reflect instantly on all devices.

Same-Day Multiple Bookings

- Multiple users can book the same slot at different times on the same day.
- Example: Slot C-01 booked from 1:00 PM to 2:00 PM and 3:00 PM to 4:00 PM.
- Prevents time collisions while maximizing slot usage throughout the day.

11. Booking Validation & Overlap Prevention

Database Time Overlap Rule

- Checks if new booking time overlaps with existing booking time.
- Overlap Condition: (NewStart < ExistEnd) AND (NewEnd > ExistStart).
- If times overlap, the database blocks the booking attempt immediately.
- Ensures zero double bookings for all vehicle types.

Clear Example

- Existing Booking: 12 August (1:00 PM to 2:00 PM).
- ALLOWED: 12 August (3:00 PM to 4:00 PM) -> No time overlap.
- BLOCKED: 12 August (1:30 PM to 2:30 PM) -> Overlaps by 30 minutes.
- Rule: Prevents double booking for the same slot at overlapping times.

12. QR-Based Check-In & Check-Out Flow

Step-by-Step Gate Workflow

- 1. Customer books slot -> QR ticket generated on phone.
- 2. Customer arrives at gate -> Admin scans QR code with camera.
- 3. Check-In: Booking status changes to 'Active / Checked In'.
- 4. Customer leaves -> Admin scans QR again to perform Check-Out.
- 5. Status changes from 'Active' to 'Past' in My Bookings.

Early Arrival & Manual Backup

- Early Check-In Block: Prevents gate entry before scheduled start time.
- Manual Plate Lookup: Admin can enter vehicle number if camera fails.
- Time Recording: Stores exact check-in and check-out times in database.

13. Notifications & Overtime Management

Real-Time Push Notifications

- Check-In Reminder: Sent 15 minutes before parking start time.
- Check-Out Reminder: Sent 15 minutes before parking end time.
- Overtime Warning: Sent if vehicle stays past the booked end time.
- Notification History: Saved in database and visible inside customer app.

Automated Overtime Calculation

- Compares Scheduled Check-Out Time with Actual Exit Time.
- Example: Scheduled 10:00 PM, Actual Exit 10:15 PM (15 mins late).
- Round Up Hours: Extra Hours = ceil(overtime_mins / 60) -> 1 hour.
- Late Fee: Overtime Fee = Extra Hours * Hourly Rate added to final bill.

14. Refund & Customer Account Management

Customer Refund Workflow

- Easy Request: Customer can request cancellation before check-in.
- Admin Review: Admin sees refund request and reason on dashboard.
- Action: Admin clicks Approve or Reject button.
- Instant Update: Customer receives push notification with refund status.

Customer Profile Editing

- Easy Profile Edit: Customer can update Full Name and Mobile Number.
- No Avatar Upload: Focused only on basic account information.
- Database Update: Updates existing record in Supabase using User ID.
- Data Sync: Updated info shows instantly in Customer App and Admin Panel.

15. AI Nearest & Available Recommendation

Distance Calculation

- Uses GPS coordinates to measure exact distance in kilometers.
- Estimates driving ETA time in minutes based on distance.

Availability Scoring

- Checks live free slots for requested vehicle type (Car, Bike, EV).
- Availability Score = Free Slots / Total Slots.
- Hides parking locations that have no empty slots available.

Simple Score & Status

- Final Score = 0.60 * Distance Score + 0.40 * Availability Score.
- Ranks top recommendations so user gets close & open parking.
- Status: Implemented via Python FastAPI microservice (Prototype).

16. System Architecture & Tech Stack

Layered System Architecture

- Frontend: Flutter mobile and web app built using Dart language.
- State Management: Provider pattern for smooth screen updates.
- Backend Database: Supabase PostgreSQL with secure access policies.
- Push Server: Firebase Cloud Messaging (FCM) for notifications.
- Python Microservice: FastAPI server for AI scoring & background timers.

External Services & Tools

- Google Maps API: Opens direct location navigation links.
- Geolocator: Detects user's GPS position for distance calculation.
- Mobile Scanner: High-speed camera QR code reader package.
- Figma & GitHub: Screen design and source code version control.

17. Database Schema & Supabase Backend

Main Database Tables

- Profiles: Stores user ID, full name, mobile number, and FCM token.
- Parking Locations: Stores location name, address, GPS coords, and price.
- Parking Slots: Stores slot code, vehicle type (Car, Bike, EV), and floor.
- Bookings: Stores user ID, slot ID, date, time range, status, and late fee.

Table Relationships

- Location contains multiple Floors and Slots.
- Slot belongs to a specific Parking Location.
- Booking connects Customer, Parking Location, and Slot.
- Row Level Security (RLS) protects user data from unauthorized access.

18. Complete System End-to-End Workflow

Customer Journey

- 1. User logs into ParkSafe app.
- 2. Searches location and views Google Maps directions.
- 3. Selects vehicle type (Car, Bike, EV) and date/time range.
- 4. Chooses floor and slot on visual grid screen.
- 5. Enters vehicle number, pays, and gets QR ticket pass.

Admin Journey

- 6. Admin scans customer QR code at entry gate.
- 7. System verifies pass and checks vehicle in.
- 8. Customer parks vehicle inside garage.
- 9. Admin scans QR code at exit gate to check out.
- 10. System checks for late fees and moves booking to Past.

19. Benefits, Future Scope & Conclusion

Main Benefits & Impact

- Saves Time: Eliminates 15-20 minutes of hunting for parking.
- Paperless: 100% digital QR code passes on mobile phone.
- Easy Management: Automated gate check-ins and overtime tracking.
- Eco-Friendly: Reduces unnecessary driving and car emissions.

Future Enhancements & Conclusion

- Camera Barrier: Automatic License Plate Recognition at gate entry.
- Smart Sensors: Ultrasonic IoT sensors for automatic slot detection.
- Dynamic Pricing: Price adjustments during high-demand peak hours.
- Summary: Complete digital solution for smart city parking.

References & Technologies Used

Flutter
Cross-platform framework for Web, Android & iOS apps

Dart
Primary programming language for mobile application

Supabase
Cloud database, user authentication & realtime backend

PostgreSQL
Relational database engine with secure data policies

Firebase FCM
Push notification service for reminders & alerts

Google Maps API
Map location links and turn-by-turn navigation

Geolocator
GPS tracking to measure distance to parking

Python & FastAPI
AI recommendation microservice REST API

APScheduler
Background worker for scheduled notifications

Figma
UI/UX screen design and logo branding assets

Antigravity
AI-assisted development workspace & code intelligence

GitHub
Version control and code repository management

PARKSAFE PROJECT PRESENTATION

Thank You!

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Questions & Evaluation

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